

Semester Senior Capstone Design Final Report

Project 1501: CNC Tube Notcher

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Objective

Develop a **program controllable, small form factor** tube notcher compatible with the existing plasma cutter at the UMass All-Campus Makerspace, accommodating steel tubes up to 1.5" in diameter and 10' in length.

Motivation

- Supermileage Vehicle (SMV) and Formula SAE (FSAE) want to create **space frames**, which require **notched tubes**
- No room at Makerspace for large conventional tube notchers

Target Specifications

Machine Size: 4'x3'x3'
Machine Weight: 50lbs
Acceleration: 0.1g
Surface Feedrate: 120 IPM
Accuracy: ±1mm/300mm, according to ISO 13920 Class A
Surface Roughness: 19.6µm Ra
Angular Backlash: 0.18 deg



Figure 1: Notched tubes prior to welding

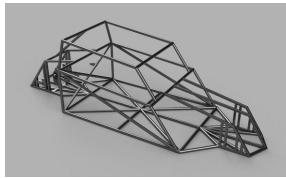


Figure 2: Space frame

Motor Sizing for Chuck

- Estimate chuck moment of inertia I
- Calculate torque balance
- Fit chuck motor to requirements

$$I = \frac{1}{2} m_c (r_{c0}^2 + r_{c1}^2) + 3m_r r_r^2 + 3m_g r_g^2 + m_p r_p^2$$

$$= 0.125 + 0.00726 + 0.00588 + 0.00163$$

$$= 0.140 \text{ [kgm}^2\text{]}$$

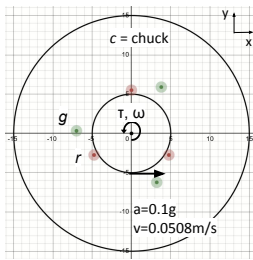


Figure 3: Chuck free body diagram

$$\tau = I\alpha \quad \alpha = \frac{a_{\text{pipe}}}{r_{\text{pipe}}}$$

$$\alpha = \frac{0.981 \text{ [m/s}^2\text{]}}{0.01905 \text{ [m]}} = 51.5 \text{ [rad/s}^2\text{]}$$

$$I = 0.140 \text{ kgm}^2$$

$$\tau = (51.5 \text{ [rad/s}^2\text{]}) (0.140 \text{ [kgm}^2\text{]})$$

$$\tau = 7.21 \text{ [Nm]}$$

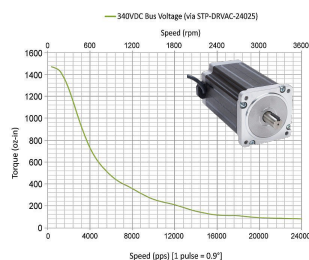


Figure 4: Motor torque curve

Chuck Bearing, Weld Analysis

$$\frac{C_0}{F_c} = \frac{6.76 \text{ kN}}{142.25 \text{ N}} = 47$$

$$\sum F_x = 0 : F_t - R_x = 0 \Rightarrow R_x = F_t$$

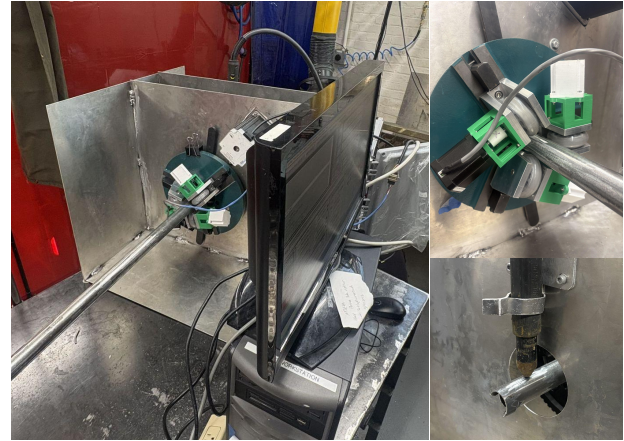
$$\sum F_y = 0 : F_r - R_y = 0 \Rightarrow R_y = F_r$$

$$\frac{4.41 \text{ N}}{142.25 \text{ N}} = 0.031 \Rightarrow F_c = F_r = 142.25 \text{ N}$$

Found **F = 5 lbs/in** of weld required:
intended throat 0.125" permissible with
weld strength reduction factor = 0.5

Design and Prototype

Dimensions: 25" x 24.5" x 30", Weight: 88.0 lbs.



Left: full assembly. Right top: chuck modifications. Right bottom: plasma head.

Wheels translate pipe

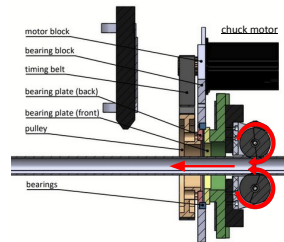


Figure 5: Section view of chuck

Chuck rotates pipe

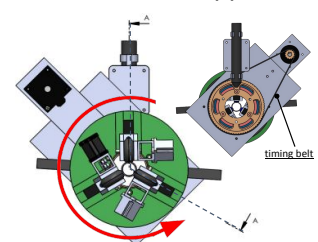


Figure 6: Front and back of chuck

Performance Evaluation Results

Particle Tracking

Take image series of isolated motion, track centroids with MATLAB

- Axial: 300 IPM, 0.1g
- Angular: 150 IPM, 0.05g
- Backlash: 1.32 deg

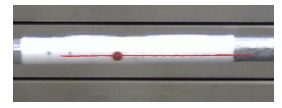
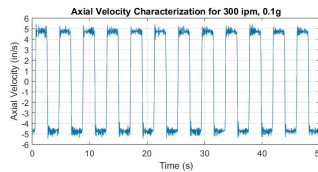


Figure 7: Tracked axial centroids of dot

Surface Roughness

Sample points with dial indicator, two sample t-test for sheet vs. tube

- 109.9µm Ra, due to cross
- p = 5.5e-5 at 95% confidence



ISO 13920: Class B angular

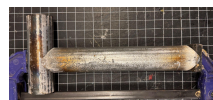
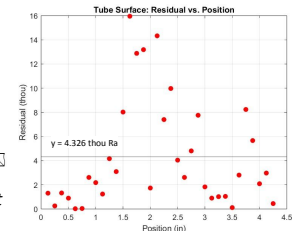
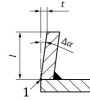


Figure 8: Testing for off-normal length t with a square and feeler gauge.



Recommendations

- Built-in steady rest instead of separate roller support
- Adjust pulley spacing to permit use of off-the-shelf timing belt
- Higher quality chuck with tighter clearances between parts